

Section 3: Reproduction and inheritance

- a) Reproduction
- b) Inheritance

a) Reproduction

Students will be assessed on their ability to:

- 3.1 understand the differences between sexual and asexual reproduction
- 3.2 understand that fertilisation involves the fusion of a male and female gamete to produce a zygote that undergoes cell division and develops into an embryo

Flowering plants

- 3.3 describe the structures of an insect-pollinated and a wind-pollinated flower and explain how each is adapted for pollination
- 3.4 understand that the growth of the pollen tube followed by fertilisation leads to seed and fruit formation
- 3.5 understand the conditions needed for seed germination**
- 3.6 understand how germinating seeds utilise food reserves until the seedling can carry out photosynthesis**
- 3.7 understand that plants can reproduce asexually by natural methods (illustrated by runners) and by artificial methods (illustrated by cuttings)

Humans

- 3.8 describe the structure and explain the function of the male and female reproductive systems
- 3.9 understand the roles of oestrogen and progesterone in the menstrual cycle
- 3.10 describe the role of the placenta in the nutrition of the developing embryo**